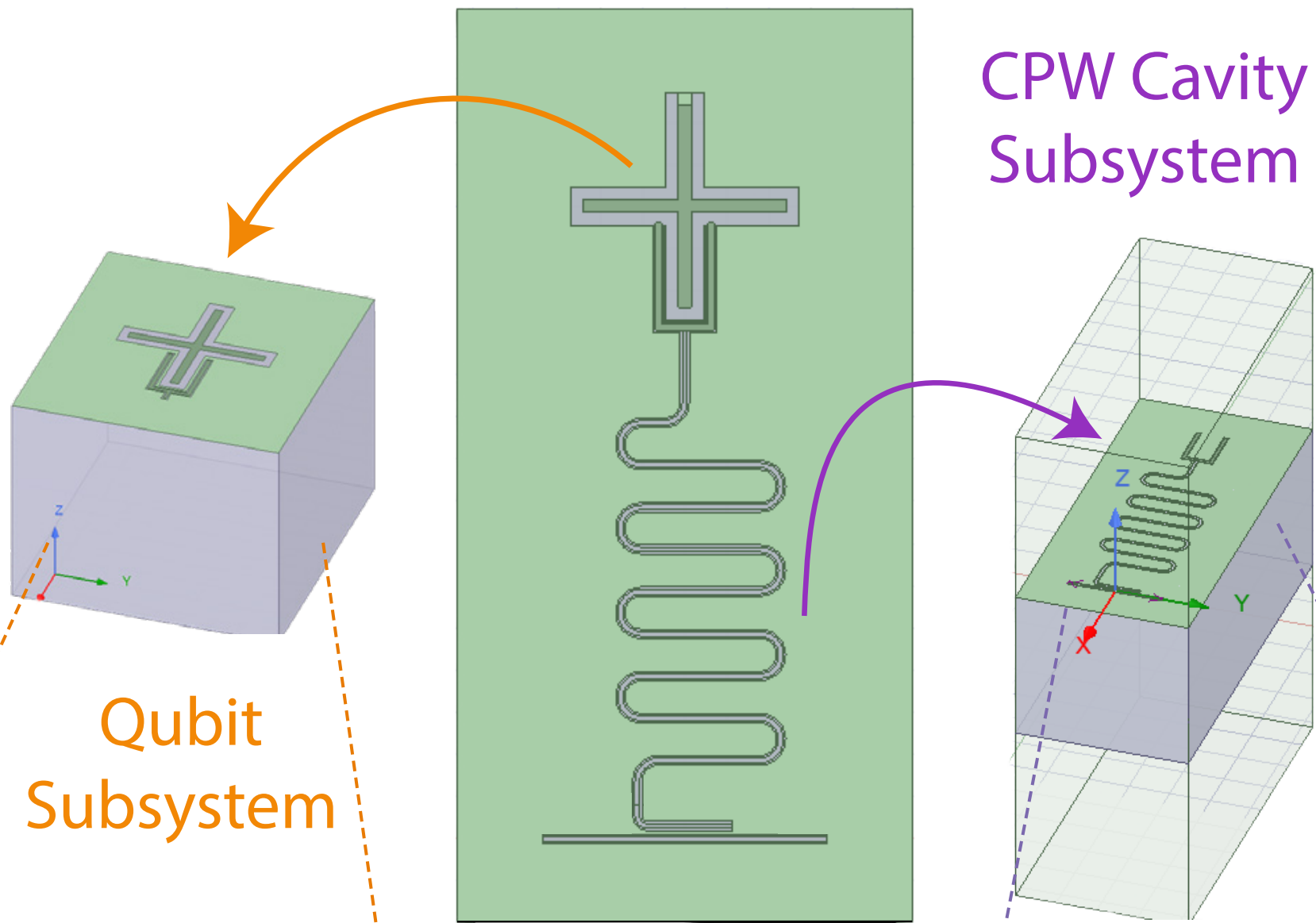




CPW Cavity
Subsystem

Qubit
Subsystem

α (anharmonicity)

$\approx -f_C$ from Q3D

f_q (qubit frequency)

$\approx \sqrt{8 f_J f_C} - f_C$ from Q3D

f_r (resonator freq.)

HFSS eigenmode solver

κ (resonator linewidth)

HFSS via 50 Ω port loss